

Public Economics

Take-Home Assignment 3

Topics: Taxation (Tax Incidence, Progressive Brackets)

Name: _____ Student ID: _____ Due: _____

Answer both questions. Where a short explanation is requested, 2–3 sentences is enough — we are grading for correct reasoning, not length. Show your work for any calculation.

Question 1: Tax Incidence

The city of Vienna is considering a per-trip tax on ride-hailing services (Uber, Bolt). The market for rides is described by the inverse supply and demand curves

$$P^S(Q) = 2 + 0.5Q, \quad P^D(Q) = 20 - Q,$$

where Q is trips per hour and P is the price in euros.

- a) Find the equilibrium price and quantity *before* any tax.
- b) The city imposes a tax of €3 per trip, collected from the **drivers** (suppliers). Find the new equilibrium quantity, the price *riders* pay, and the price *drivers* keep after remitting the tax. How much of the €3 tax is borne by riders, and how much by drivers? In one sentence, connect your answer to the relative steepness (elasticity) of supply and demand in this market.

Question 2: Progressive Brackets, Average vs. Marginal Rates

Austria's (simplified) income tax schedule has the following brackets:

Taxable income	Rate
Up to €13,539	0%
€13,539 – €21,992	20%
€21,992 – €36,458	30%
€36,458 – €70,365	40%
€70,365 – €104,859	48%

Markus has taxable income of €50,000.

- a) Calculate Markus's total tax bill by working up through the brackets. What is his **marginal** tax rate, and what is his **average** tax rate?
- b) Next year, Markus's nominal income rises by 5% (to €52,500), but prices have also risen by 5%, so his *real* purchasing power is exactly unchanged. If the brackets above stay fixed in nominal terms, explain in 2–3 sentences what happens to Markus's average tax rate, and name the phenomenon this illustrates. Briefly state how Austria's actual tax policy since 2023 addresses it.